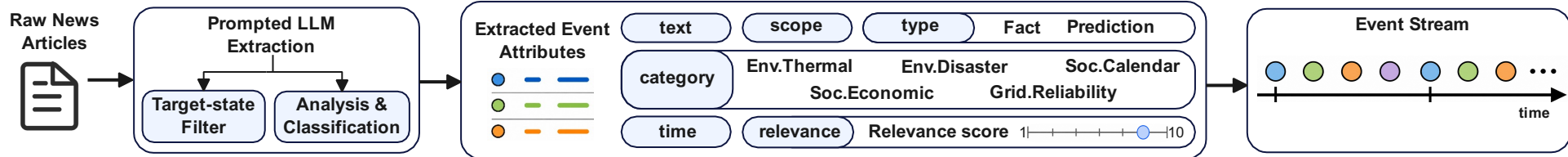
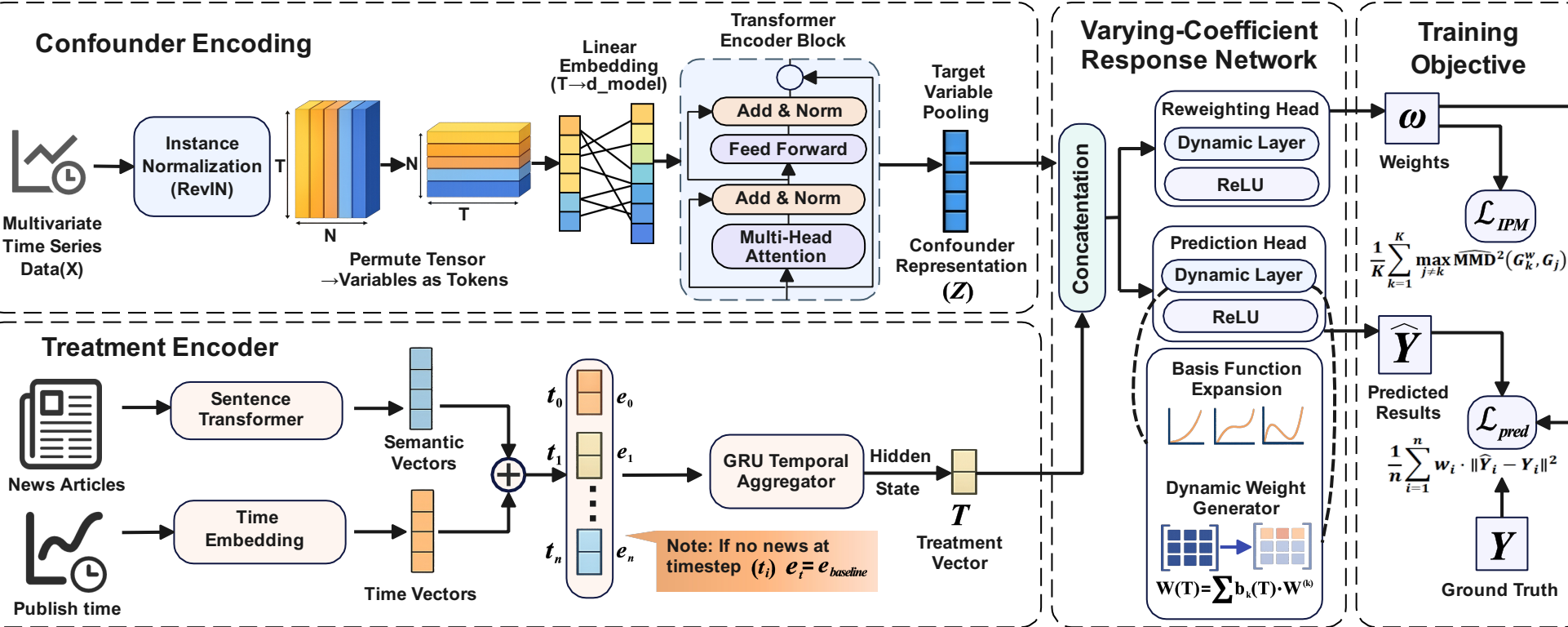


1.LLM-based Structured Event Stream Constructions



2.Model Architecture



3.Counterfactual Inference and Load Analysis

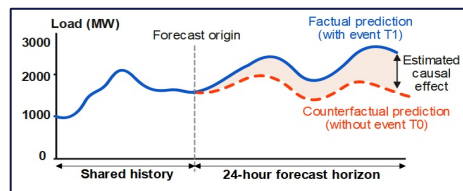
Estimates and analyzes news impact

Factual Load Forecast

$$\hat{Y}_{\text{factual}} = f_{\text{pred}}(Z, T)$$

No-news Counterfactual Forecast:

$$\hat{Y}_{\text{counterfactual}} = f_{\text{pred}}(Z, T_0)$$



- ◇ Perturbation trajectory over prediction horizon
- ◇ ITE / ATE / CATE
- ◇ Category-level news effects
- ◇ Event-level case study interpretation
- ◇ Load analysis for grid operators